Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name: Xcore**
b. **Team leader name: Thanveer Ahamed**
c. **Problem Statement:** Intelligent Risk, Fraud & Financial Experiences (BFSI). Lenders and insurers judge property risk from slow site visits or self-reported photos nobody can verify. We turn a 20-second phone video into a fraud-checked, explainable, underwriting-ready property risk file.

Property Inspection AI

Phone video → underwriting-ready property risk file



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Leader · Gen AI



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Vision AI Lead



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Solution Architect

Brief about the idea

From a 20-second phone video to an underwriting-ready property risk file

Property Inspection AI is a Gen AI inspection service for mortgage lenders, insurers and inspectors. Upload a phone walkthrough video; in minutes it returns defects, condition grade A–E, perils, repair cost, valuation impact, premium loading, an underwriting decision with reasons, fraud flags and a 3D model that proves what was inspected.

THE PROBLEM: property risk is judged from photos nobody can verify

Slow site visits	Unverifiable photos	No proof of coverage	Subjective grading	Overvaluation
Days of waiting; remote cases go unchecked	Reused, edited, off-screen or wrong property	Which walls did the inspection see?	Inspectors disagree; reasons unrecorded	RBI has flagged inflated collateral values

WHO IT SERVES: one inspection record, five BFSI decision-makers

Lenders, HFCs, NBFCs	Insurers: underwriting	Insurers: claims	Inspectors & valuers	Owners & buyers
Collateral grade, valuation impact, REFER flag	Perils, risk level, premium loading	Repair-cost range, fraud-checked evidence	AI-drafted report, re-film list	Condition report, 3D model, floor plan

$308.6B
yearly US insurance fraud, all lines (Coalition Against Insurance Fraud, 2022)

~$45B
of that in property & casualty insurance (same study)

$4.5–12.5B
global home-inspection services market, 2025 (industry reports)



Minutes, not days · Fraud-resistant · Explainable

Concept illustration of the product

Opportunities

How it is different, how it solves the problem, and our USP

HOW DIFFERENT IS IT FROM EXISTING IDEAS?

	Site visit / photo-only AI tools	Property Inspection AI
Turnaround	Days (visit) / minutes (tools)	**Minutes; ~10 min per 20 s video**
Proof of what was inspected	Inspector's word / none	**10 cm coverage map + 3D-pinned findings**
Evidence fraud checks	None / limited	**Keyframe dHash reuse, Gemini screen / AI check**
How the decision is made	Subjective / black-box score	**Transparent YAML rules; Gemini explains**
False-positive control	Human / single model	**Find (LocateAnything) → verify (Gemma 4) → rules**
BFSI outputs & integration	Report or damage list, PDF	**Grade, perils, cost, valuation, premium; BigQuery + Pub/Sub**

HOW WILL IT SOLVE THE PROBLEM?

Slow site visits	→	full risk file in minutes; low-risk homes clear without a visit
Unverifiable photos	→	every walkthrough fraud-checked; any HIGH flag → REFER
No proof of coverage	→	< 60% AI-inspected surface → REFER with a re-film list
Subjective grading	→	score = 100 − Σ(points × severity × state weight)
Overvaluation	→	valuation-impact %, repair retention, REFER before disbursal

USP of the proposed solution

Proof of inspection. A 10 cm coverage map and 3D-pinned findings show exactly what the AI saw, and what to film again.

Find → verify → decide. Two models cut false positives; deterministic rules make every decision auditable.

Fraud-resistant evidence. Recycled videos, screen recordings and AI footage are flagged before a decision.

One record, five teams. Lending, underwriting, claims, inspectors and owners share one BigQuery / Pub/Sub record.

About $1 per video. ≈ $0.35 of Gemini per walkthrough; one-command Cloud Run deploy that scales to zero.

List of features offered by the solution

One upload gives a complete, explainable risk file

1. **Defects localised & verified**
   18 defect types on 12 components
2. **Condition grade A–E**
   0–100 score, transparent formula
3. **Risk level & insurance perils**
   Water, structural, moisture...
4. **Indicative repair-cost range**
   Per finding and per property
5. **Valuation & collateral impact**
   Impact % and repair retention
6. **Premium-loading suggestion**
   For quote and renewal
7. **Underwriting decision + reasons**
   PROCEED / CONDITIONS / REFER
8. **Evidence-integrity flags**
   Recycled video, screens, AI footage
9. **Video → 3D, floor plan, coverage**
   GLB models, room areas, maps
10. **Gemini analysis & Q&A**
    Grounded answers citing issue IDs

✓ **Detection scope**

Walls, ceilings, floor & wall tiles, grout, wood/vinyl, countertops, doors, locks, windows, trim, kitchen exhausts in 8 room types. Look-alikes (grout, grain, shadows) rejected; locks, windows, fans always manual; cracks get a structural note, water/mold a moisture check.

✓ **Fraud & evidence integrity**

Keyframe dHash reuse across every property, Gemini screen / render / AI check on every keyframe, blurred frames dropped, video coverage < 60% → REFER. Any HIGH flag → human review.

✓ **3D digital twin & coverage map**

Depth Anything 3 + OpenCV 5 SLAM + Open3D TSDF → clean & SLAM GLB, floor plan (PNG + SVG), navigation map. Every 10 cm of surface checked; greedy AI keyframes; findings pinned to a wall and video time.

✓ **Gemini co-pilot & product**

Property analysis, grounded Q&A, scene and object naming (explains, never changes results). Portfolio dashboard, live 3D-scan view, Lite / Advanced 3D, 3D viewer, HTML / Markdown / JSON reports, REST API at /docs.

Process flow diagram or Use-case diagram

Capture, inspect, decide, act: minutes instead of days

ACTORS

- **Inspector / owner** captures
- **Underwriter / claims** reviews & decides
- **Core systems** LOS, policy, claims

1 Capture
A phone walkthrough video. No special hardware.

2 AI inspection
Gemini reads the room; LocateAnything finds defects; Gemma 4 verifies each crop.

3 Risk decision
Rules set grade, perils, cost, valuation, premium, decision; Gemini explains.

4 Act
Report, Gemini Q&A, BigQuery row and Pub/Sub event to core systems.

DECISION OUTCOMES (back to underwriter and core systems)

- PROCEED
- PROCEED W/ CONDITIONS: note defects
- PROCEED W/ CONDITIONS: repairs / retention
- REFER: professional survey
- REFER: evidence integrity
- REFER: additional evidence

KEYFRAME PIPELINE

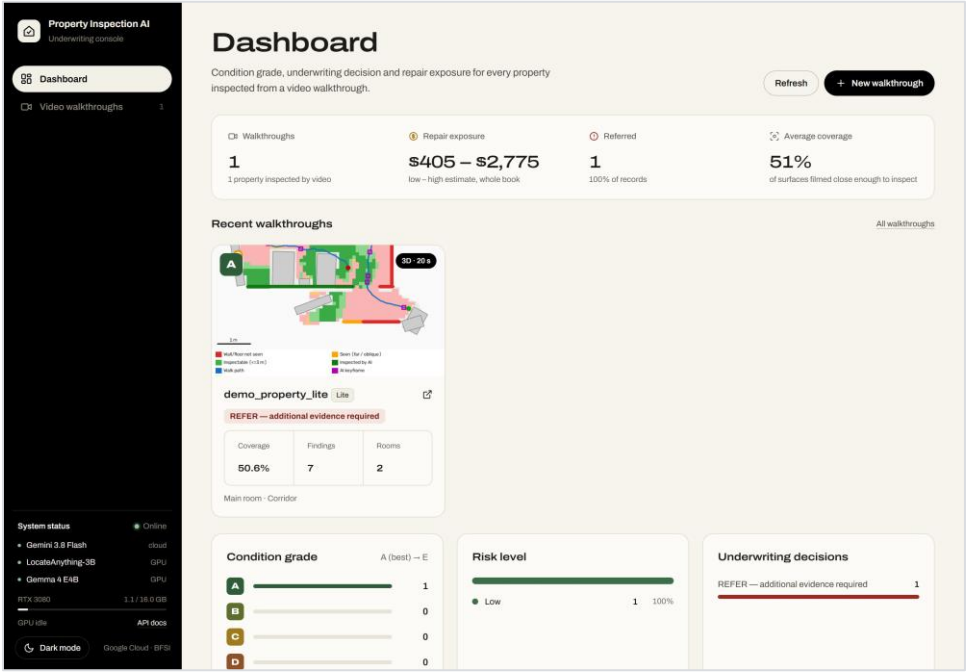
1. Quality + integrity (dHash)
2. Gemini: room, components, authenticity
3. Prompt router: 18 defects, 12 components
4. LocateAnything-3B boxes → NMS
5. Full-res crops → Gemma 4 verify
6. Rule engine → status → risk engine

VIDEO PIPELINE

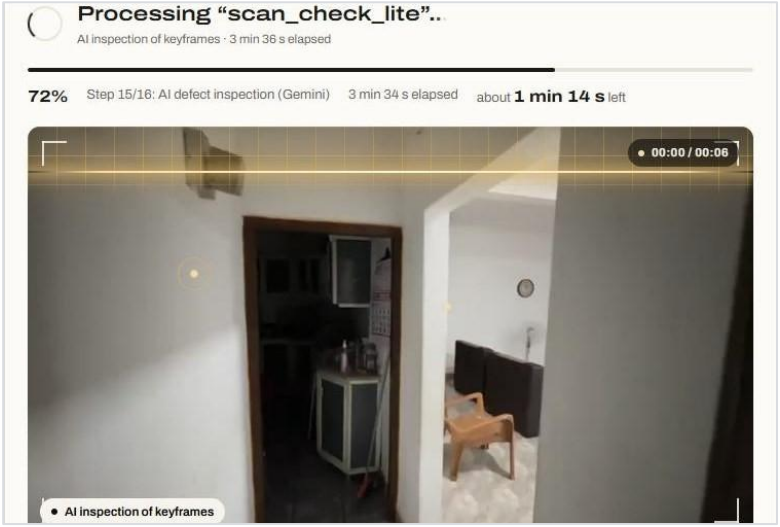
1. Depth Anything 3 depth per frame
2. OpenCV 5 SLAM camera poses
3. Open3D TSDF surface fusion
4. Walls, floor, doors; Gemini names objects
5. 10 cm coverage + AI keyframes
6. Keyframes → keyframe pipeline; pins in 3D

Wireframes/Mock diagrams of the proposed solution (optional)

Actual product screens: an inspection app built for underwriters



Dashboard: walkthroughs, repair exposure, referrals, coverage and condition grades across the book

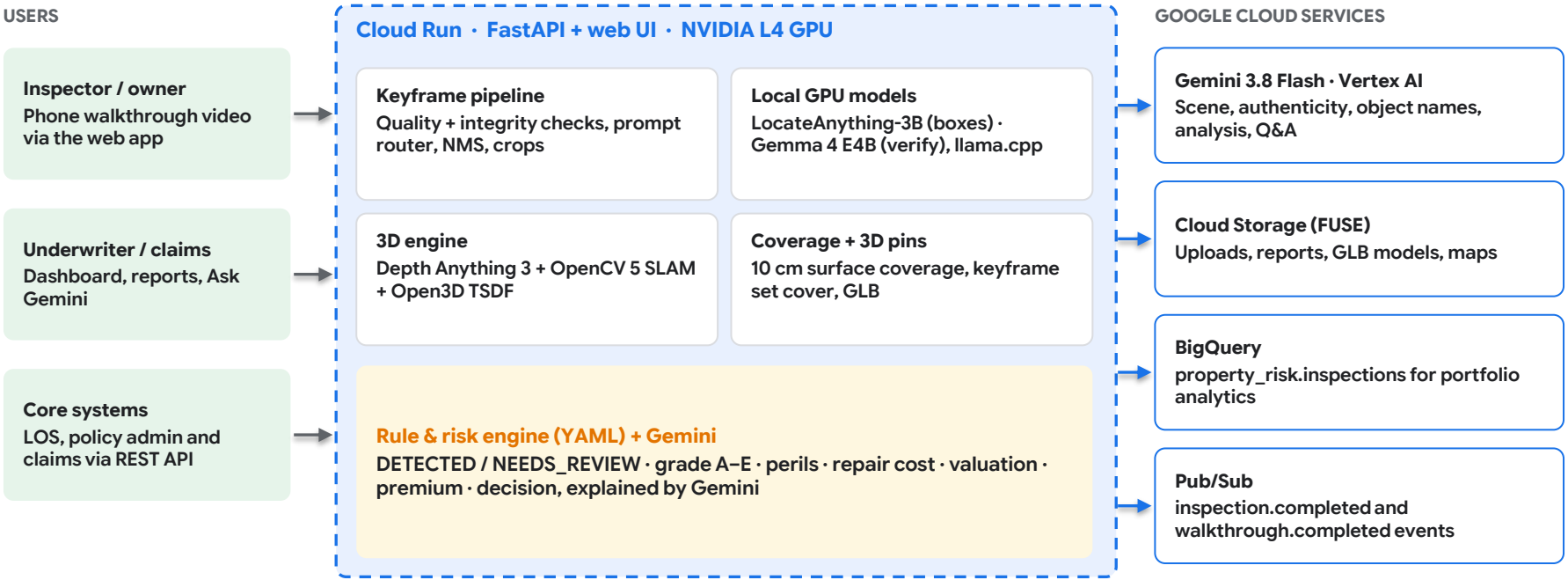


Video walkthrough: live 3D-scan progress while the video is processed

- Dashboard → Video walkthroughs → 3D report → Ask Gemini
- Video upload with a live 3D-scan view, Lite or Advanced
- Interactive 3D viewer: walk inside, see coverage gaps
- Reports in HTML, Markdown, JSON; REST API at /docs

Architecture diagram of the proposed solution

One Cloud Run service, powered by Gemini on Vertex AI



Build & ops: Cloud Build + Artifact Registry · least-privilege IAM service account (no API keys in production) · Cloud Logging · one-command deploy.sh · scales to zero

Technologies to be used in the solution

Built on Google AI and Google Cloud, with best-in-class open models

GOOGLE AI & CLOUD, IN THE PROTOTYPE

Gemini on Vertex AI
Scene, authenticity, analysis, grounded Q&A, 3D object names

Gemma 4 (E4B)
Open Google model that verifies every defect crop on the GPU

Cloud Run + NVIDIA L4
The whole app in one container; scales to zero

Cloud Storage
FUSE-mounted data: uploads, reports, 3D models

BigQuery
One row per inspection: grade, decision, perils, flags, coverage

Pub/Sub
Event-driven hand-off to underwriting and claims

Cloud Build + Artifact Registry
Reproducible container build with baked-in models

IAM service account + Logging
Least privilege, no keys in production

OPEN MODELS & LIBRARIES

LocateAnything-3B open-vocabulary defect boxes

llama.cpp local GPU inference for Gemma 4

Depth Anything 3 monocular depth per frame

OpenCV 5 SLAM tracking, ICP, CV evidence

Open3D TSDF fusion, meshes, GLB export

gsplat Gaussian-splatting experiment

FastAPI + Python 18 REST endpoints, web UI, /docs

YAML rule engine client-owned risk rules

NEXT ON THE ROADMAP

Document AI
Title deeds, valuation reports, repair invoices cross-checked with images

Vertex AI Search
Policy wording and lending guidelines for the assistant

Vertex AI + BigQuery ML
Claim-likelihood and collateral-risk models

Cloud SQL / AlloyDB
Multi-tenant case management, audit trail, SSO

Why this stack: Gemini does reasoning, authenticity and explanation; open models on the Cloud Run GPU do high-volume boxing and verification cheaply; deterministic rules keep decisions auditable. Data stays in the client's own Google Cloud project, and BigQuery + Pub/Sub plug straight into lending, underwriting and claims systems.

Estimated implementation cost (optional)

Priced per inspection, far below a physical site visit

UNIT ECONOMICS (INDICATIVE, FROM THE PROTOTYPE)

Cost item	Basis	Estimate
Gemini calls (Vertex AI)	≈ $0.005 per call	≈ $0.35 per walkthrough
NVIDIA L4 GPU on Cloud Run	≈ $1.42 per hour (8 vCPU, 32 GiB)	Scales to zero when idle
Video inspection compute	10–35 min of GPU + Gemini	≈ $1 Lite, ≈ $1.50 Advanced
Two weeks of judging	~40 visits, 20 videos	≈ $50–70 in total
Storage, BigQuery, Pub/Sub	Pay-per-use managed services	Billed by usage
Deployment & ops	One-command deploy.sh, Cloud Build	No servers to manage

≈ $0.35
Gemini per walkthrough

< $1
GPU compute per video

$0
GPU cost when idle (scales to zero)

BUSINESS MODEL

Pay-per-inspection API
Lenders and insurers send a walkthrough video from their own apps; get JSON, a report and a decision back.

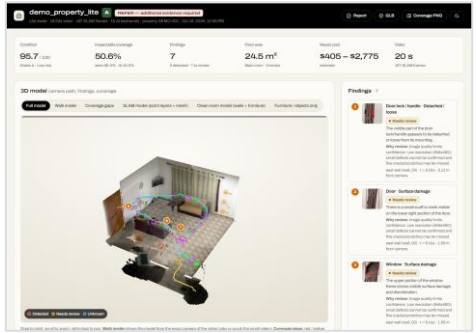
Enterprise subscription
Portfolio dashboard, BigQuery export, custom risk rules, SSO, private deployment in the client's GCP project.

Inspector & valuer firms
White-label app: AI drafts every report and coverage control, so each inspector handles more cases.

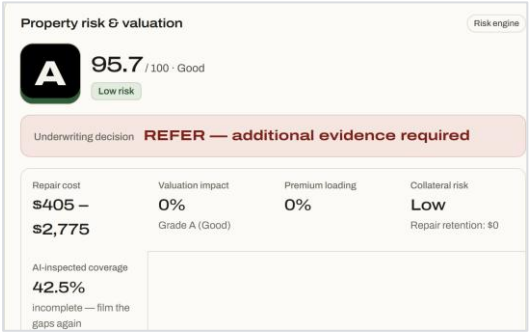
Pricing hypothesis to validate in pilots: a per-inspection fee well below a physical visit, with volume tiers for portfolios.

Snapshots of the prototype

Real screens and outputs from the working prototype v1.0.0



3D report: findings, grade A, decision



Risk & valuation panel: REFER with reasons



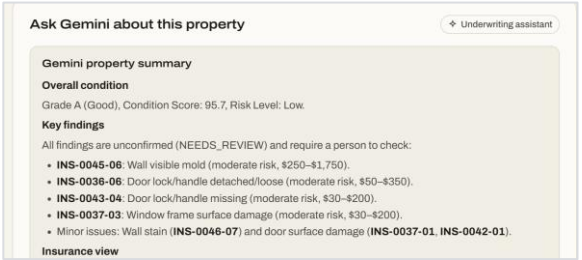
Keyframe detection: door and window damage



Phone video → clean 3D model (GLB)



Coverage map: what to film again



Gemini property analysis (co-pilot)

Prototype Performance report/Benchmarking

Measured on a real 20-second walkthrough, plus a fair 4-pipeline evaluation

CASE STUDY: REAL HOME WALKTHROUGH (MEASURED)

19.6 s phone video	**197** SLAM frames	**15** AI keyframes
24.5 m² 2 rooms measured	**95.7 · A** condition score	**$405–2,775** indicative repairs

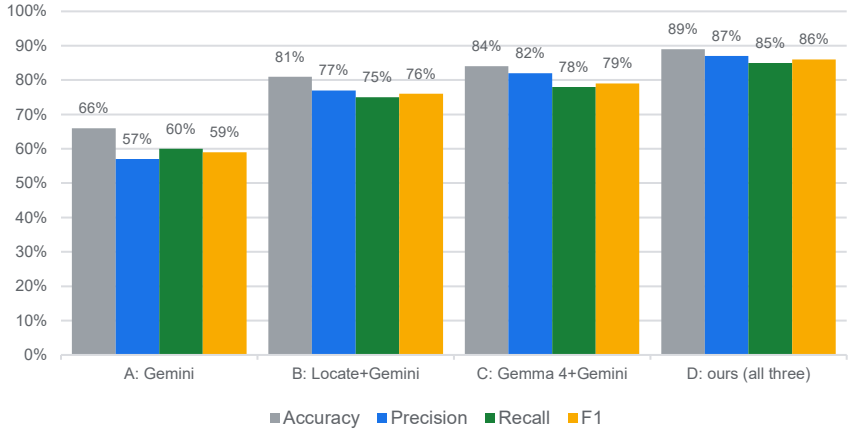
Decision: REFER, additional evidence required

Only 42.5% of surfaces were AI-inspected (min 60%), so the app lists the walls and ceilings to film again. 7 possible defects pinned in 3D (Needs review).

SPEED & COST (FROM THE PROTOTYPE)

10.6 min end to end, 20 s video, laptop GPU	**11.1 / 26.9 min** 3D build, Lite / Advanced
21.8 / 21.6 dB PSNR, Lite / Advanced	**≈ $0.35** Gemini per walkthrough

EVALUATION: 4 PIPELINES · 10 ROOMS · 100 REGIONS



Simulated illustration, not yet measured: values show the report format and the expected gain; they will be replaced after running the protocol on D.

Fair test: same images, schema, Gemini settings and decision rule; test set frozen; 5 living rooms + 5 bedrooms incl. 2 clean controls; 40 defective / 60 clean regions.

Targets for D: ≥ 85% accuracy, precision, recall, F1; ≤ 1 false alarm on clean rooms; beat A on every metric.

Additional Details/Future Development (if any)

Roadmap: documents, policies and predictive portfolio risk

NOW · v1.0.0

- Video walkthrough inspection
- Risk and valuation engine
- Fraud and coverage checks
- Gemini analysis and Q&A
- Cloud Run, BigQuery, Pub/Sub

NEXT · 3 months

- Mobile guided capture with live coverage
- Document AI: deeds, valuations, invoices
- Vertex AI Search over policy wording
- Commercial localiser (Gemini / Vertex-tuned)

LATER · 6–12 months

- BigQuery ML claim & collateral-risk scoring
- Before/after claim comparison in 3D
- Multilingual reports (Gemini)
- AlloyDB multi-tenant case management

WHAT WE BUILT: FROM DEPTH RESEARCH TO A DEPLOYABLE PRODUCT

01 Depth + SLAM
Depth Anything 3 visual SLAM, occupancy grid, A* path

02 Hybrid SLAM
OpenCV 5 tracking, ICP, Manhattan planes

03 Photo-real 3D
Gaussian Splatting experiment (gsplat)

04 Production 3D
Video → GLB models, floor plan (+SVG)

05 Inspection AI
Taxonomy, locate + verify, risk engine, fraud

06 Google Cloud v1.0.0
Gemini, Cloud Run + L4, BigQuery, Pub/Sub

18
defect types, 12 components

8
room types

18
REST API endpoints

2
3D modes (Lite / Advanced)

1
command to deploy on GCP

Additional Details/Future Development (contd.)

Responsible AI, go-to-market and the team behind it

RESPONSIBLE AI: HUMANS DECIDE, AI SHOWS EVIDENCE

1 Human in the loop
Decision support only; a qualified surveyor or underwriter makes the final call.

2 Explainable
Deterministic YAML rules; every decision lists its reasons and risk points.

3 Auditable
Raw outputs archived; each inspection a BigQuery record; versioned config.

4 Safe wording
No structural-safety or root-cause claims; locks, fans, windows stay manual.

5 Secure
Service account, no API keys in files, data stays in the client's project.

6 Licence-aware
LocateAnything is non-commercial; Gemini localisation is the drop-in swap.

GO-TO-MARKET: HOUSING FINANCE & HOME INSURANCE, APAC

PILOT · 0–3 months
1 housing-finance lender + 1 home insurer; AI alongside existing inspections on ~500 cases.

INTEGRATE · 3–6 months
Pub/Sub into loan-origination and claims; BigQuery dashboards; custom rules per client.

SCALE · 6–18 months
Google Cloud Marketplace, valuer & inspector networks, commercial property and claims.

Pilot KPIs: turnaround · cost per inspection · % auto-cleared vs referred · fraud flags confirmed · re-inspection rate

"Every lender and insurer already asks for photos. We replace them with a short phone video that is trustworthy, explainable and ready for a decision."

Team Xcore: Thanveer Ahamed (Team Leader, Gen AI: Gemini & Gemma reasoning, verification, underwriting assistant) · Janitha Prathapa (Vision AI: defect detection, Depth Anything 3, SLAM, 3D, coverage map) · Dinal Fernando (Solution Architect: Cloud Run, Vertex AI, BigQuery, Pub/Sub, deployment)

Provide links to your:

- GitHub Public Repository: github.com/XCore-Lab/property-inspection-ai
- Demo Video Link (3 Minutes): [paste link here]
- Final Product Link: [paste link here]



Property Inspection AI · Prototype v1.0.0 · one-command Cloud Run deploy

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

